OneDrive for Business Deployment & Governance

Implementation Solution Guide

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Contents

Executive Summary — OneDrive for Business Deployment & Governance 4
- The outcome . . . 4
- The approach . . . 4
- Why it's low-risk . . . 4

1. Solution Overview — OneDrive for Business Deployment & Governance 5
- What this solution is . . . 5
- Why this matters . . . 5
- How it works . . . 5
- The reference design . . . 6
- What you can control . . . 7
- Limitations (be honest with buyers) . . . 7
- Naming convention (used in every doc and script) . . . 8
- The standard deployment model . . . 8

2. Requirements — OneDrive for Business Deployment & Governance 9
- 1. A managed device fleet with a client-policy channel (Intune or Group Policy) . . . 9
- 2. Devices Entra joined / hybrid joined for silent sign-in . . . 9
- 3. Tenant ID and Active Directory domain GUID(s) captured . . . 9
- 4. Storage-quota and retention policy decided . . . 10
- 5. Org-level SharePoint sharing baseline known (sharing alignment ceiling) . . . 10
- 6. Offboarding process and delegate model agreed . . . 10

Global Requirements — SharePoint Online & OneDrive Solutions 11
- 1. Microsoft 365 tenant with SharePoint Online / OneDrive and licensing . . . 11
- 2. Administrative access (least-privilege SharePoint roles) . . . 11
- 3. SharePoint Online Management Shell / PnP + Graph toolchain . . . 12
- 4. Information architecture & identity/group source of truth . . . 12
- 5. A pilot scope and change control . . . 12
- 6. Security, compliance & records stakeholders . . . 12

3. Implementation — OneDrive for Business Deployment & Governance 13
- Stage 0 — Assess (read-only baseline) . . . 13
- Stage 1 — Design (name the baseline) . . . 13
- Stage 2 — Configure the client-deployment policy (pilot ring) . . . 14
- Stage 3 — Configure Known Folder Move (pilot ring) . . . 15
- Stage 4 — Configure sync controls (tenant — validate on pilot devices) . . . 15
- Stage 5 — Configure storage, retention, and sharing (tenant) . . . 16
- Stage 6 — Configure offboarding & data-retention (validate on a test departure) . . . 16

Stage 7 — Validate the pilot . . . 17
Stage 8 — Expand tenant-wide . . . 17
Rollback (documented, per high-impact change) . . . 17
5. Testing & Validation — OneDrive for Business Deployment & Governance 18
Test 1 — Silent sign-in (`OneDrive Silent Config – AllUsers – Prod`) . . . 18
Test 2 — Known Folder Move (`KFM Silent Redirect – Pilot – Stage1`) . . . 18
Test 3 — Domain-join sync restriction (`Sync Restriction – DomainJoined – Prod`) . . . 18
Test 4 — Personal-account block (`DisablePersonalSync`) . . . 18
Test 5 — Storage quota & retention (`OneDrive Storage Quota – Default – Prod`, `OneDrive Retention – Orphaned – Prod`) . . . 19
Test 6 — Sharing alignment (`OneDrive Sharing – Aligned – Prod`) . . . 19
Test 7 — Offboarding access (`Offboarding Access – Manager – Prod`) . . . 19
Test 8 — Governance report (`OneDrive Governance – Tenant – Report`) . . . 19
Go / No-Go gate (before Stage 8 expansion) . . . 19
6. Troubleshooting — OneDrive for Business Deployment & Governance 20
Silent sign-in doesn't happen (client shows "Sign in") . . . 20
KFM didn't redirect (folders still on the local profile) . . . 20
Sync isn't blocked on an unmanaged device (restriction not working) . . . 21
Sync is blocked on a device that should be allowed (over-blocked) . . . 21
Personal account still syncs (personal-sync block not working) . . . 21
Offboarding — can't access the departed user's OneDrive . . . 21
Governance report is empty or partial . . . 22
`Connect-SPOService` fails (wrong admin URL / cloud) . . . 22
7. Best Practices — OneDrive for Business Deployment & Governance 22
Deployment & client . . . 22
Sync governance . . . 22
Storage, retention & sharing . . . 23
Offboarding & lifecycle . . . 23
Operate & assure . . . 23
8. Deliverables Summary — OneDrive for Business Deployment & Governance 24
Documentation . . . 24
PowerShell scripts (5) — safe by default (read-only or `-WhatIf`) . . . 25
Appendix A — PowerShell Scripts 27

Executive Summary — OneDrive for Business Deployment & Governance

The outcome

User files protected, synced, and governed. OneDrive for Business is deployed **silently** to every managed device, Desktop/Documents/Pictures are **redirected into OneDrive** (Known Folder Move) so a lost or reimaged laptop no longer means lost data, sync is **restricted to managed devices**, storage and retention are set to governed values, sharing is **aligned to policy**, and when an employee **leaves**, their files are preserved and handed to their manager — not orphaned or destroyed.

The approach

The solution follows a disciplined, pilot-first model — **Assess** → **Design** → **Configure** → **Pilot** → **Validate** → **Expand** — using only Microsoft-native tooling: the Microsoft 365 and SharePoint admin centers, the OneDrive client policies (Intune or Group Policy ADMX), the SharePoint Online Management Shell / PnP PowerShell, and Microsoft Graph. It configures the full OneDrive lifecycle in one coherent pass:

- **Silent client deployment** — the sync client signs users in with their Windows credentials, no prompts.
- **Known Folder Move** — Desktop, Documents, and Pictures move into OneDrive automatically, opt-out blocked.
- **Sync controls** — sync limited to domain-joined devices, personal accounts blocked, risky file types excluded, and (optionally) unmanaged browser sessions restricted to web-only.
- **Storage, retention & sharing** — a governed default quota, a deliberate orphaned-OneDrive retention window, and OneDrive sharing aligned to (never exceeding) the organization's sharing policy.
- **Offboarding** — a manager-access-and-retention step so departing users' data is retrievable, with legal/records holds deferred to compliance.

Five PowerShell tools — discovery, a `-WhatIf` configuration helper, a `-WhatIf` offboarding helper, a governance report, and a scope/identifier helper — make the work repeatable and auditable.

Why it's low-risk

- **Pilot-first with explicit gates.** Every high-impact change (sync, sharing, retention, KFM) is validated on a named pilot ring against pass/fail tests before any tenant-wide rollout, and each change has a documented rollback.
- **Safe-by-default automation.** Every script is read-only or requires an explicit `-Commit`/`-WhatIf` preview; none stores credentials or secrets.
- **Native and reversible.** No third-party tooling; all settings are standard Microsoft controls that can be reverted to the recorded baseline.
- **Compliance stays with compliance.** Retention and legal-hold **determinations** are deferred to the

organization's records and legal functions — the solution provides the technical method and evidence, not the determination.

The result is the difference between OneDrive as an uncontrolled liability and OneDrive as a governed, protected, and cost-managed part of the estate — delivered self-guided, tested, at a fraction of consulting cost.

1. Solution Overview — OneDrive for Business Deployment & Governance

What this solution is

OneDrive for Business is where most users' working files actually live. Deployed carelessly it becomes a liability: the sync client is signed in by hand (or not at all), the Desktop, Documents, and Pictures folders stay on the local disk and vanish with a lost or reimaged laptop, personal OneDrive accounts sync corporate data onto unmanaged machines, storage grows without limit, sharing drifts wider than policy allows, and when an employee leaves nobody can retrieve their files. This solution is a repeatable, pilot-first way to **deploy** the OneDrive sync client silently and **govern** it end to end — client deployment and silent sign-in, Known Folder Move (KFM), sync controls, storage and retention, sharing alignment, and offboarding/data-retention on departure — using only the Microsoft 365 admin center, the SharePoint admin center, the OneDrive client policies (Intune or Group Policy ADMX), the SharePoint Online Management Shell / PnP PowerShell, and Microsoft Graph. No third-party tooling.

Why this matters

- **Data loss is the default.** Until KFM redirects Desktop/Documents/Pictures into OneDrive, those folders live only on the device. A lost, stolen, failed, or wiped machine takes the files with it.
- **Silent deployment is the difference between adoption and shelfware.** If users must find, install, and sign in to the client themselves, coverage is partial and inconsistent; silent sign-in with Windows credentials makes OneDrive present and working on every managed device with zero user steps.
- **Uncontrolled sync is a data-exfiltration and support problem.** Without controls, users sync personal OneDrive accounts on corporate machines, sync corporate data onto unmanaged personal machines, and sync file types that should never leave (e.g. PST). Sync controls close those paths.
- **Sharing and storage drift.** OneDrive sharing can quietly be more permissive than the org intends, and unbounded quotas hide cost and stale data. Both need a governed baseline.
- **Offboarding without a plan destroys or orphans data.** When an account is deleted the OneDrive is retained only for a configured window, then permanently removed. Without a manager-access and retention step, a departure means either lost files or a scramble.

How it works

1. **Deploy the client silently.** The OneDrive sync client ships with Windows and Microsoft 365 Apps;

the solution ensures a current, machine-wide install and enables **silent account configuration** (`SilentAccountConfig`) so the client signs the user in with their Windows/Entra credentials — no prompts. Deployed via **Intune** (Settings Catalog / Administrative Templates — ADMX-backed) or **Group Policy** (the OneDrive `OneDrive.admx`/`.adml` in the Central Store).

1. **Redirect known folders (KFM).** The **Silently move Windows known folders to OneDrive**

(`KFMSilentOptIn`, keyed to your **tenant ID**) policy moves Desktop, Documents, and Pictures into OneDrive with no user interaction; **Prevent users from moving their known folders back to the PC** (`KFMBlockOptOut`) keeps them protected. A notification variant informs users when redirection completes.

1. **Control sync.** In the **SharePoint admin center** → **Settings** → **OneDrive** → **Sync** (and the matching

`Set-SPOTenantSyncClientRestriction` cmdlet) the tenant restricts sync to **domain-joined computers** (an allow-list of **AD domain GUIDs**), blocks specific file extensions, and optionally blocks Mac sync. **Prevent users from syncing personal OneDrive accounts** (`DisablePersonalSync`) blocks personal accounts on managed devices. Tenant-level **allow/block organization** lists (`AllowTenantList` / `BlockTenantList`, keyed by **tenant GUID**) stop the client from signing in to other organizations' tenants. Browser **web-only / limited access** (`ConditionalAccessPolicy = AllowLimitedAccess`) blocks download, print, and sync from unmanaged sessions.

1. **Set storage and retention.** A **default OneDrive storage quota**

(`Set-SPOTenant -OneDriveStorageQuota`, per-site override `Set-SPOSite -StorageQuota`) and the **orphaned-OneDrive retention period** (`OrphanedPersonalSitesRetentionPeriod`, 30–3650 days — how long a deleted user's OneDrive survives) are set to governed values.

1. **Align sharing.** `OneDriveSharingCapability` is set no more permissive than the org-level

SharePoint sharing setting (OneDrive can never exceed it), so OneDrive sharing matches policy.

1. **Govern offboarding.** On departure, a delegate (typically the **manager**) is granted

site-collection admin on the leaver's OneDrive to retrieve needed files, retention preserves the data for the configured window, and any legal/records **hold** is applied by compliance (Purview) before deletion — the technical steps here, the determination there.

1. **Pilot first, then expand.** Every change is validated on a named pilot ring before tenant-wide

rollout, with explicit pass/fail gates and a documented rollback.

The reference design

Layer	What it does	Where it's set
Client deployment	Machine-wide, current-ring sync client, silent sign-in	Intune / GPO ADMX (`SilentAccountConfig`, `FilesOnDemandEnabled`)
Known Folder Move	Redirect Desktop/Documents/Pictures into OneDrive, block opt-out	Intune / GPO ADMX (`KFMSilentOptIn` = tenant ID, `KFMBlockOptOut`)

Layer	What it does	Where it's set
Sync controls	Domain-joined only, block extensions, block personal accounts, org allow/block	SPO admin center Sync + `Set-SPOTenantSyncClientRestriction`, `DisablePersonalSync`, `AllowTenantList/BlockTenantList`
Browser access	Web-only (blocks download/print/sync) for unmanaged sessions	`Set-SPOTenant -ConditionalAccessPolicy AllowLimitedAccess`
Storage & retention	Default quota + per-user override; orphaned-site retention	`Set-SPOTenant -OneDriveStorageQuota` / `-OrphanedPersonalSitesRetentionPeriod`
Sharing	OneDrive sharing aligned to (≤) org sharing level	`Set-SPOTenant -OneDriveSharingCapability`
Offboarding	Manager access grant + retention; hold deferred to compliance	`Set-SPOUser -IsSiteCollectionAdmin`; Purview hold

What you can control

- The **deployment channel** (Intune vs Group Policy) and the pilot vs broad **client-config rings**.
- Whether KFM is **silent**, silent-with-notification, or wizard-prompted, and whether opt-out is blocked.
- The **sync restrictions**: which domain GUIDs may sync, which file extensions are blocked, whether personal accounts and Mac sync are allowed, and which organizations the client may sign in to.
- The **default quota** and any **per-user overrides**, and the **orphaned-OneDrive retention** window.
- The **OneDrive sharing** capability (kept at or below the org sharing level).
- The **offboarding** delegate and handoff process.

Limitations (be honest with buyers)

- **KFM requires the folders be eligible.** Known Folder Move silently moves Desktop/Documents/Pictures only; folders that are redirected elsewhere by legacy Folder Redirection, are on a network path, or contain unsupported/locked content can block the move and must be remediated first (see Troubleshooting).
- **Silent sign-in requires the device be joined** (Entra joined / hybrid joined) with the user's primary account; on workgroup or personal devices `SilentAccountConfig` cannot complete and the user must sign in interactively.
- **Sync controls are client policies**, enforced by the OneDrive client on managed devices — they govern the sync client, not browser access; browser download/print is governed separately by conditional-access limited access.

- **Retention is a deletion timer, not a backup.** The orphaned-OneDrive retention period governs how long a deleted user's OneDrive is kept; it is not versioned backup and does not substitute for a legal hold, which is a **compliance/records determination** (deferred to Purview and legal).
- **Plan- and cloud-dependent behavior.** Default storage limits, some sharing controls, and admin URLs differ between plans and between **Commercial and Government** clouds (e.g. `*-admin.sharepoint.us`). These are flagged **"verify in your tenant"** throughout — never invented.

Naming convention (used in every doc and script)

Every artifact this solution creates is named `<Purpose> – <Scope> – <Stage>` and reused verbatim across the docs, scripts, admin centers, and Intune/GPO. The standard set:

Artifact	Name
Pilot user ring	`OneDrive Pilot Ring – Users – Stage1`
Broad user ring	`OneDrive Broad Ring – Users – Stage2`
Silent client-config profile (pilot)	`OneDrive Client Config – Pilot – Stage1`
Silent client-config profile (broad)	`OneDrive Client Config – Broad – Stage2`
Silent sign-in setting	`OneDrive Silent Config – AllUsers – Prod`
KFM redirect (pilot)	`KFM Silent Redirect – Pilot – Stage1`
KFM redirect (broad)	`KFM Silent Redirect – Broad – Stage2`
Sync domain-join restriction	`Sync Restriction – DomainJoined – Prod`
Org sync allow list	`Sync Tenant Restriction – OrgOnly – Prod`
Default storage quota	`OneDrive Storage Quota – Default – Prod`
Orphaned-site retention	`OneDrive Retention – Orphaned – Prod`
OneDrive sharing alignment	`OneDrive Sharing – Aligned – Prod`
Offboarding manager access	`Offboarding Access – Manager – Prod`
Discovery baseline export	`OneDrive Baseline – Tenant – Discovery`
Governance report	`OneDrive Governance – Tenant – Report`

The standard deployment model

> *Assess the current OneDrive posture → design the client-config, KFM, sync, storage, sharing, and offboarding baseline → configure it → roll to a **pilot** ring first → validate against explicit gates → expand tenant-wide, keeping a documented rollback for every high-impact change. Pattern: **Assess** → **Design** → **Configure** → **Pilot** → **Validate** → **Expand.***

2. Requirements — OneDrive for Business Deployment & Governance

These are the **solution-specific** prerequisites for deploying and governing OneDrive for Business. They build **on top of** the shared SharePoint-category prerequisites in `global-requirements-sharepoint.md` — read that first. It already covers the six shared items: (1) a Microsoft 365 tenant with SharePoint Online / OneDrive and the right licensing, (2) least-privilege SharePoint administrative access, (3) the SPO Management Shell / PnP + Graph toolchain, (4) the identity/group source of truth, (5) a pilot scope and change control, and (6) security/compliance/records stakeholders. Do **not** re-verify those here; verify the OneDrive-specific items below.

Each requirement is stated as **Description / Why / Verify / Implement if missing**.

1. A managed device fleet with a client-policy channel (Intune or Group Policy)

• **Description:** A population of managed Windows devices reachable by **Microsoft Intune** (Settings Catalog / Administrative Templates — ADMX-backed) **or** on-prem **Group Policy** with the current OneDrive `OneDrive.admx`**/**`.adml` in the domain Central Store. macOS devices, if in scope, are governed by the equivalent OneDrive `.plist` configuration profile.

• **Why required:** Silent sign-in, KFM, Files On-Demand, and sync restrictions are **client policies** delivered to the device — without a policy channel they cannot be enforced at scale.

• **Verify:** Confirm devices are Intune-enrolled (or domain-joined with GPO applying) and that the OneDrive ADMX is present in the Central Store (`\\<domain>\SYSVOL\<domain>\Policies\PolicyDefinitions\OneDrive.admx`), or that the OneDrive Administrative Templates appear in Intune's Settings Catalog.

• **Implement if missing:** Enroll devices in Intune, or download the current OneDrive ADMX/ADML from the sync client install folder (`%localappdata%\Microsoft\OneDrive\<version>\adm\`) into the Central Store.

2. Devices Entra joined / hybrid joined for silent sign-in

• **Description:** In-scope devices are **Microsoft Entra joined** or **hybrid Entra joined** with the user signed in under their corporate account.

• **Why required:** `SilentAccountConfig` signs the user in with their Windows/Entra credentials; it can only complete when the device carries a usable primary work account. On workgroup or personal devices the user must sign in interactively.

• **Verify:** `dsregcmd /status` shows `AzureAdJoined : YES` (or `DomainJoined : YES` + `AzureAdJoined : YES` for hybrid) on a sample device.

• **Implement if missing:** Bring devices into Entra join / hybrid join before relying on silent sign-in.

3. Tenant ID and Active Directory domain GUID(s) captured

• **Description:** The **Microsoft 365 tenant ID** (a GUID) and, where domain-join sync restriction is used, the **on-prem AD domain GUID(s)**.

• **Why required:** The KFM **Silently move known folders** policy is keyed to the **tenant ID**, and **Allow syncing only on computers joined to specific domains** is keyed to **domain GUIDs** — the wrong GUID silently no-ops the policy.

• **Verify:** Tenant ID from the Entra admin center (Overview) or `ODFBGOV-Set-Scope.ps1 -TenantOnlyLookup; domain GUID from (Get-ADDomain).ObjectGUID or ODFBGOV-Set-Scope.ps1 -ResolveDomainGuids`.

• **Implement if missing:** Retrieve and record both before authoring the client policies.

4. Storage-quota and retention policy decided

• **Description:** A decided **default OneDrive storage quota** (and any per-role overrides) and the **orphaned-OneDrive retention period** (30–3650 days) for departed users.

• **Why required:** Both are governed baselines; the retention period is also the **hard deletion timer** for a leaver's OneDrive, so it must be set deliberately with compliance input, not left at default.

• **Verify:** Current values via `Get-SPOTenant` (`OneDriveStorageQuota`, `OrphanedPersonalSitesRetentionPeriod`); confirm the target values are agreed with storage/records owners.

• **Implement if missing:** Agree the target quota and retention window with the storage and compliance/records stakeholders (GR item 6) before configuring.

5. Org-level SharePoint sharing baseline known (sharing alignment ceiling)

• **Description:** The tenant's **org-level SharePoint external sharing** setting, which is the ceiling OneDrive sharing cannot exceed.

• **Why required:** `OneDriveSharingCapability` can be set equal to or **more restrictive** than the org SharePoint setting, never more permissive — so the org baseline must be known to align OneDrive.

• **Verify:** `Get-SPOTenant | Select SharingCapability, OneDriveSharingCapability`.

• **Implement if missing:** Confirm the org sharing baseline (often set by the External Sharing solution) before aligning OneDrive.

6. Offboarding process and delegate model agreed

• **Description:** An agreed **offboarding** process naming who receives access to a departing user's OneDrive (typically the **manager**) and for how long, and where legal/records **holds** are decided.

• **Why required:** The offboarding step grants operational access and relies on retention to preserve

data; the **hold determination** is a compliance/records decision that must be owned before an account is deleted.

• **Verify:** Confirm the HR/IT joiner-mover-leaver process references a OneDrive handoff step and a compliance checkpoint.

• **Implement if missing:** Define the delegate model and the compliance hold checkpoint with HR, records, and legal before running offboarding at scale.

*Compliance and legal determinations — retention schedules, legal-hold decisions, and records classification of OneDrive content — are **deferred** to the organization's compliance, records, and legal functions. This solution provides the technical method (retention period, manager access, evidence) and defers the determination.*

Global Requirements — SharePoint Online & OneDrive Solutions

These requirements apply to **every** solution in the **SharePoint** category (SharePoint Online architecture & provisioning, external sharing, OneDrive for Business, migration to SPO, permissions, search, and records/retention). Verify each before starting a solution's implementation steps.

> ***Vendor boundary (applies to every SharePoint solution).*** *The **SharePoint admin center**, the **Microsoft 365 admin center**, the **SharePoint Online Management Shell** and **PnP PowerShell**, and **Microsoft Graph** are authored at exact, current click-by-click / cmdlet detail from Microsoft's public documentation. Where a capability, setting, or portal path is **plan- or version-dependent** (storage limits, sharing controls, Syntex/records features, Commercial vs Government cloud), it is **cited or flagged "verify in your tenant"** — never invented. **Third-party migration and governance tools** (e.g. ShareGate, Migration Manager, AvePoint) are integrated at the pattern level; their consoles are deferred to the vendor's docs. Automation is **PowerShell** (SPO Management Shell / PnP / Graph) — never with hardcoded credentials or secrets.*

1. Microsoft 365 tenant with SharePoint Online / OneDrive and licensing

• **Description:** A Microsoft 365 tenant with **SharePoint Online** and **OneDrive for Business** and the licensing the capability requires (e.g. an E-plan for advanced records/retention, Syntex where used).

• **Why required:** All site, sharing, OneDrive, and records configuration depends on the right plan.

• **How to verify:** In the M365 admin center confirm the SharePoint/OneDrive subscription and add-ons.

• **How to implement if missing:** Assign/purchase the required licensing before starting.

2. Administrative access (least-privilege SharePoint roles)

• **Description:** Access with the least-privilege role the solution names — typically **SharePoint Administrator** (and Global Administrator only where genuinely required, e.g. tenant-wide sharing).

• **Why required:** Tenant/site settings, sharing, provisioning, and records changes require the role.

- **How to verify:** Confirm the role in the M365 admin center / Entra roles.
- **How to implement if missing:** Have a Global Admin assign the least-privilege role under change control.

3. SharePoint Online Management Shell / PnP + Graph toolchain

- **Description:** A workstation with the **SharePoint Online Management Shell** and/or **PnP PowerShell** (and Graph PowerShell where used), modern-auth / MFA-capable admin sign-in (PnP app registration where required).
- **Why required:** Configuration, automation, and the provided scripts run through SPO/PnP/Graph.
- **How to verify:** `Connect-SPOService` / `Connect-PnPOnline` succeeds; modules installed.
- **How to implement if missing:** Install the modules; register the PnP app where needed; confirm sign-in.

4. Information architecture & identity/group source of truth

- **Description:** A defined **information-architecture** intent (hub/site structure, taxonomy) and the identity/group source (Entra ID groups, M365 Groups backing team sites) that drives access.
- **Why required:** Provisioning, permissions, search, and governance key off the IA and groups.
- **How to verify:** Confirm the intended hub/site structure and which groups model is authoritative.
- **How to implement if missing:** Define the IA and group model (often the Architecture solution) first.

5. A pilot scope and change control

- **Description:** A **pilot site collection / hub / user set** to validate on before tenant-wide rollout, and a **change-control** process for production changes (sharing, permissions, retention, migration cutover are high-impact).
- **Why required:** Every solution follows an **Assess** → **Design** → **Configure** → **Pilot** → **Validate** → **Expand** model; tenant-wide sharing/permission/retention changes are high-impact.
- **How to verify:** Confirm a named pilot scope and an approval/change window are available.
- **How to implement if missing:** Create a pilot scope and agree change control before production changes.

6. Security, compliance & records stakeholders

- **Description:** Engagement with the **security, compliance/records, and data-governance** stakeholders whose policies the configuration must satisfy (external sharing, permissions, retention/records, sensitivity).
- **Why required:** Sharing, permission, and records settings must align to enterprise policy.

- **How to verify:** Confirm the applicable policies and named approvers.
- **How to implement if missing:** Identify the owners and capture requirements before configuring governance.

*The deployment model across every SharePoint solution is the same: **Assess** → **Design** → **Configure** → **Pilot** → **Validate** → **Expand**, with explicit pass/fail gates and a documented rollback for production changes. Automation is delivered as **PowerShell** (SPO Management Shell / PnP / Graph) — never with hardcoded credentials or secrets in code.*

3. Implementation — OneDrive for Business Deployment & Governance

Follow these steps in order. They implement the reference design from the Overview using the **Assess** → **Design** → **Configure** → **Pilot** → **Validate** → **Expand** model. Every artifact is named with the `<Purpose> – <Scope> – <Stage>` convention and reused verbatim in Testing, Troubleshooting, and the scripts. Tenant-wide sync, sharing, retention, and KFM changes are **high-impact** — pilot first, gate every stage, and keep the documented rollback ready.

> ***Cloud note.*** *Portal paths use Commercial cloud URLs (`*-admin.sharepoint.com`). For **Government** clouds substitute the Government URLs (e.g. `*-admin.sharepoint.us`) — **verify in your tenant**. Do not assume a Commercial path exists unchanged in GCC/GCC High.*

Stage 0 — Assess (read-only baseline)

Step 0.1 — Capture the current OneDrive posture. Run the discovery script read-only and save the baseline `OneDrive Baseline – Tenant – Discovery`:

```
.\ODFBGOV-Discovery.ps1 -AdminUrl https://contoso-admin.sharepoint.com `
    -CsvPath C:\ODFBGOV\OneDrive-Baseline.csv -IncludeGraphUsage
```

Record: default quota, orphaned-site retention, `OneDriveSharingCapability` vs org `SharingCapability`, current sync-client restriction state, provisioned OneDrive count, and top storage consumers.

Step 0.2 — Resolve the identifiers. Capture the tenant ID and domain GUID(s) for the KFM and sync policies:

```
.\ODFBGOV-Set-Scope.ps1 -TenantOnlyLookup      # tenant ID (KFM + tenant sync restriction)
.\ODFBGOV-Set-Scope.ps1 -ResolveDomainGuids    # AD domain GUID(s) (domain-join sync restriction)
```

Gate 0 (pass/fail): Baseline captured, tenant ID and domain GUID(s) recorded, and the deployment channel (Intune or GPO) confirmed present (Requirement 1). **Fail** → resolve before proceeding.

Stage 1 — Design (name the baseline)

Step 1.1 — Define the rings. In Entra ID create/confirm two groups: `OneDrive Pilot Ring – Users – Stage1` (a small, representative set including at least one KFM-eligible and one edge-case device) and `OneDrive Broad Ring – Users – Stage2` (the remaining population).

Step 1.2 — Decide the target values and record them in the design (each becomes a named artifact):

• `OneDrive Storage Quota – Default – Prod` — default quota in MB (e.g. 1048576 = 1 TB; **verify your

plan's max** — plan-dependent).

• `OneDrive Retention – Orphaned – Prod` — orphaned-OneDrive retention days (30–3650), agreed with records.

• `OneDrive Sharing – Aligned – Prod` — target `OneDriveSharingCapability` (≤ org `SharingCapability`).

• `Sync Restriction – DomainJoined – Prod` — the domain GUID allow-list + blocked file extensions.

• `Sync Tenant Restriction – OrgOnly – Prod` — the tenant-ID allow list (own tenant only).

• KFM mode: silent (`KFM Silent Redirect – …`), block opt-out yes/no, notification yes/no.

Gate 1: Rings created, every target value decided and approved through change control. **Fail** → do not configure until approved.

Stage 2 — Configure the client-deployment policy (pilot ring)

Configure to the **pilot ring only** first. Steps below give the **Intune** path with the equivalent **Group Policy** setting named; use whichever channel is your standard (Requirement 1).

Step 2.1 — Create the client-config profile `OneDrive Client Config – Pilot – Stage1`**.** Intune admin center → **Devices** → **Configuration** → **Create** → **Windows 10 and later** → **Settings catalog**. Name it `OneDrive Client Config – Pilot – Stage1`. Add the OneDrive settings:

Setting (Intune / GP name)	Value	Policy value
Silently sign in users to the OneDrive sync app with their Windows credentials	Enabled	`SilentAccountConfig = 1` (`OneDrive Silent Config – AllUsers – Prod`)
Use OneDrive Files On-Demand	Enabled	`FilesOnDemandEnabled = 1`
Prevent users from syncing personal OneDrive accounts	Enabled	`DisablePersonalSync = 1`

Step 2.2 — Ensure the sync client is current and machine-wide. OneDrive ships in Windows and Microsoft 365 Apps. Confirm the pilot devices run a current build; for machine-wide (per-machine) install where standalone deployment is needed, install with the documented per-machine switch (`OneDriveSetup.exe` with the all-users install parameter) — **defer exact packaging to your Intune/Configuration Manager app model.**

Step 2.3 — Assign `OneDrive Client Config – Pilot – Stage1` to `OneDrive Pilot Ring – Users – Stage1` only.

Gate 2: Profile assigned; on a pilot device `SilentAccountConfig` is present (registry `HKLM\SOFTWARE\Policies\Microsoft\OneDrive`) and the client signs in with no prompt. **Fail** → Troubleshooting (silent sign-in).

Stage 3 — Configure Known Folder Move (pilot ring)

Step 3.1 — Add KFM to the pilot profile `OneDrive Client Config – Pilot – Stage1` (artifact `KFM Silent Redirect – Pilot – Stage1`):

Setting (Intune / GP name)	Value	Policy value
Silently move Windows known folders to OneDrive	Enabled, **Tenant ID = your tenant GUID**	`KFMSilentOptIn = <tenant ID>`
Prevent users from moving their Windows known folders to their PC	Enabled	`KFMBlockOptOut = 1`
(optional) **Silently move … with notification**	Enabled to notify users when redirect completes	`KFMSilentOptInWithNotification = 1`

> *The tenant ID **must** be the GUID from Step 0.2. A wrong or blank tenant ID makes the policy no-op silently.*

Step 3.2 — Deploy to the pilot ring and allow the client to redirect Desktop, Documents, and Pictures into OneDrive.

Gate 3: On a pilot device the three known folders show the OneDrive redirect (path under `%UserProfile%\OneDrive - <Org>\`), files are present in OneDrive online, and opt-out is greyed out. **Fail** → Troubleshooting (KFM blocked).

Stage 4 — Configure sync controls (tenant — validate on pilot devices)

Sync controls are **tenant-level**; scope validation to pilot devices before broad reliance.

Step 4.1 — Domain-join sync restriction `Sync Restriction – DomainJoined – Prod`**.** SharePoint admin center → **Settings** → **OneDrive** → **Sync** → **Allow syncing only on computers joined to specific domains** → paste the **domain GUID(s)** from Step 0.2. In this pane also set **Block syncing of specific file types** (e.g. `pst`) and, if required, **Block Mac sync**. PowerShell equivalent (preview first with `ODFBGOV-Configure.ps1`):

```
# Preview:
.\ODFBGOV-Configure.ps1 -AdminUrl https://contoso-admin.sharepoint.com `
    -DomainGuids '12345678-90ab-cdef-1234-567890abcdef' -WhatIf
# Apply:
.\ODFBGOV-Configure.ps1 -AdminUrl https://contoso-admin.sharepoint.com `
    -DomainGuids '12345678-90ab-cdef-1234-567890abcdef' -Commit
# Direct cmdlet equivalent:
Set-SPOTenantSyncClientRestriction -Enable -DomainGuids "12345678-90ab-cdef-1234-567890abcdef" -ExcludedFi
```

Step 4.2 — Organization sync restriction `Sync Tenant Restriction – OrgOnly – Prod`**.** To stop the client signing in to other organizations' tenants, allow only your own tenant ID (`AllowTenantList`) via the **Allow syncing OneDrive accounts for only specific organizations** policy (or block specific tenants with **Block syncing OneDrive accounts for specific organizations**, `BlockTenantList`). These are **client ADMX policies keyed by tenant GUID** — deploy through the same Intune/GPO channel; **verify the exact value names against your OneDrive ADMX build**.

Step 4.3 — Browser web-only access (optional). To block download/print/sync from unmanaged browser sessions, set limited access (requires the conditional-access prerequisites — **verify licensing**):

```
.\ODFBGOV-Configure.ps1 -AdminUrl https://contoso-admin.sharepoint.com -ConditionalAccess AllowLimitedAcce
# equivalent: Set-SPOTenant -ConditionalAccessPolicy AllowLimitedAccess
```

Gate 4: On a **non-domain-joined** test device the OneDrive client refuses to sync (restriction works); on a pilot **domain-joined** device sync succeeds; blocked extensions are refused. **Fail** → Troubleshooting (sync blocked / not blocked).

Stage 5 — Configure storage, retention, and sharing (tenant)

Step 5.1 — Storage quota + retention + sharing. Preview then apply the named artifacts:

```
# Preview all three:
.\ODFBGOV-Configure.ps1 -AdminUrl https://contoso-admin.sharepoint.com `
    -StorageQuotaMB 1048576 -RetentionDays 365 -OneDriveSharing ExistingExternalUserSharingOnly -WhatIf
# Apply:
.\ODFBGOV-Configure.ps1 -AdminUrl https://contoso-admin.sharepoint.com `
    -StorageQuotaMB 1048576 -RetentionDays 365 -OneDriveSharing ExistingExternalUserSharingOnly -Commit
```

This sets `OneDrive Storage Quota – Default – Prod` (`-OneDriveStorageQuota`), `OneDrive Retention – Orphaned – Prod` (`-OrphanedPersonalSitesRetentionPeriod`), and `OneDrive Sharing – Aligned – Prod` (`-OneDriveSharingCapability`, kept ≤ org `SharingCapability`). Admin-center equivalents: **SharePoint admin center** → **Settings** → **OneDrive storage limit** and → **Sharing**.

Step 5.2 — Per-user quota overrides (as needed). `Set-SPOSite -Identity <OneDriveUrl> -StorageQuota <MB>` for roles that justify more/less than the default.

Gate 5: `Get-SPOTenant` shows the target quota, retention, and OneDrive sharing values, and OneDrive sharing is **not** more permissive than org sharing. **Fail** → re-apply / align.

Stage 6 — Configure offboarding & data-retention (validate on a test departure)

Step 6.1 — Manager access `Offboarding Access – Manager – Prod`**.** For a departing user, grant the delegate (manager) access to the leaver's OneDrive so files can be retrieved before deletion:

```
# Preview:
.\ODFBGOV-Set-OffboardingAccess.ps1 -AdminUrl https://contoso-admin.sharepoint.com `
    -DepartingUser jdoe@contoso.com -Delegate manager@contoso.com -WhatIf
# Apply:
.\ODFBGOV-Set-OffboardingAccess.ps1 -AdminUrl https://contoso-admin.sharepoint.com `
    -DepartingUser jdoe@contoso.com -Delegate manager@contoso.com -Commit
```

The tenant retention window (`OneDrive Retention – Orphaned – Prod`) preserves the OneDrive for the configured days after the account is deleted. The SharePoint admin center can also **automatically grant the manager access** for a set number of days on deletion — configure that under **Settings** → **OneDrive** → **Retention/Notifications** to match `Offboarding Access – Manager – Prod`.

Step 6.2 — Hold before deletion (compliance-owned). If the leaver is subject to any legal/records **hold**, compliance applies a Purview retention/eDiscovery hold **before** the account is deleted. This is a **compliance/legal determination — deferred**; IT provides the technical retention window and

manager-access evidence, compliance decides and applies the hold.

Gate 6: On a test departure the manager can open the OneDrive and retrieve files, and the retention period is confirmed. **Fail** → Troubleshooting (offboarding access).

Stage 7 — Validate the pilot

Run the full validation before expanding (see **05-testing.md**), then re-run discovery to confirm the posture and generate the governance report:

```
.\ODFBGOV-Discovery.ps1 -AdminUrl https://contoso-admin.sharepoint.com -IncludeGraphUsage   # confirm tena
.\ODFBGOV-Report.ps1 -AdminUrl https://contoso-admin.sharepoint.com -HtmlPath C:\ODFBGOV\OneDrive-Report.h
```

Client-side gates (silent sign-in, KFM redirect, sync enforcement) are verified **on the pilot device** per the procedures in **05-testing.md** (registry checks + observed client behavior).

Gate 7 (go/no-go): All pilot gates green — silent sign-in, KFM redirect, sync controls enforced, storage/retention/sharing correct, offboarding works. **Fail any** → hold expansion, remediate, re-test.

Stage 8 — Expand tenant-wide

Step 8.1 — Broad client config. Clone the pilot profile to `OneDrive Client Config – Broad – Stage2` (silent sign-in + Files On-Demand + KFM `KFM Silent Redirect – Broad – Stage2` + personal-sync block) and assign to `OneDrive Broad Ring – Users – Stage2`.

Step 8.2 — Confirm tenant settings (sync restriction, quota, retention, sharing) are already tenant-wide from Stages 4–5; no per-ring change needed.

Step 8.3 — Monitor with `ODFBGOV-Report.ps1` on a recurring cadence (near-quota and stale accounts).

Gate 8: Broad ring reports silent sign-in + KFM at target coverage; governance report clean.

Rollback (documented, per high-impact change)

- **Silent sign-in / KFM:** unassign the client-config profile (or set `KFMBlockOptOut = 0` and remove

`KFMSilentOptIn`); redirection already completed stays in OneDrive — reverse per Microsoft KFM guidance.

- **Sync restriction:** `Set-SPOTenantSyncClientRestriction -Disable` (re-run `ODFBGOV-Configure.ps1`

without `-DomainGuids` does not disable — use the direct cmdlet) to lift the domain-join allow list.

- **Sharing / quota / retention:** re-run `ODFBGOV-Configure.ps1 -Commit` with the prior baseline values

recorded in Step 0.1.

- **Offboarding grant:** `Set-SPOUser -Site <url> -LoginName <delegate> -IsSiteCollectionAdmin $false`.

5. Testing & Validation — OneDrive for Business Deployment & Governance

Run these after each stage and as the go/no-go gate before tenant-wide expansion. Each test has an explicit **expected pass** and a **fail** signal. Use the named artifacts from the Implementation guide. Client-side tests run **on a pilot device** signed in as a pilot-ring user; tenant tests run from the SPO Management Shell.

Test 1 — Silent sign-in (`OneDrive Silent Config – AllUsers – Prod`)

Procedure (pilot device): After the profile `OneDrive Client Config – Pilot – Stage1` applies, sign in as a pilot user on a freshly-synced device. Confirm the policy landed:

```
Get-ItemProperty 'HKLM:\SOFTWARE\Policies\Microsoft\OneDrive' -Name SilentAccountConfig
```

Expected pass: `SilentAccountConfig = 1`, and the OneDrive client is signed in with the user's work account **without any prompt** (cloud icon in the system tray, "Up to date"). **Fail:** value absent/0, or the client shows "Sign in" — see Troubleshooting (silent sign-in).

Test 2 — Known Folder Move (`KFM Silent Redirect – Pilot – Stage1`)

Procedure (pilot device):

```
Get-ItemProperty 'HKLM:\SOFTWARE\Policies\Microsoft\OneDrive' -Name KFMSilentOptIn, KFMBlockOptOut
```

Then open **File Explorer** → **Desktop / Documents / Pictures** and check the path. **Expected pass:** `KFMSilentOptIn` equals the **tenant ID**, `KFMBlockOptOut = 1`; the three folders resolve under `%UserProfile%\OneDrive - <Org>\` and their contents appear in OneDrive online; the "Stop protecting" / move-back option is greyed out. **Fail:** folders still on the local profile path, or opt-out available — see Troubleshooting (KFM blocked).

Test 3 — Domain-join sync restriction (`Sync Restriction – DomainJoined – Prod`)

Procedure: Confirm the tenant setting, then test client behavior on two devices:

```
Get-SPOTenantSyncClientRestriction | Select TenantRestrictionEnabled, AllowedDomainList, ExcludedFileExten
```

- On a **domain-joined / allowed-GUID** pilot device: sync a file.
- On a **non-domain-joined** test device: attempt to sync.

Expected pass: `TenantRestrictionEnabled = True` with the correct domain GUID(s); allowed device syncs; non-allowed device is blocked with the "your organization does not allow syncing on this computer" message; a blocked extension (e.g. `.pst`) is refused. **Fail:** restriction not enabled, or a non-allowed device syncs — see Troubleshooting (sync not blocked).

Test 4 — Personal-account block (`DisablePersonalSync`)

Procedure (pilot device): In the OneDrive client, attempt to add a **personal** Microsoft account.

```
Get-ItemProperty 'HKLM:\SOFTWARE\Policies\Microsoft\OneDrive' -Name DisablePersonalSync
```

Expected pass: `DisablePersonalSync = 1`; the client refuses to add a personal OneDrive account. **Fail:** personal account adds successfully — re-check the policy assignment.

Test 5 — Storage quota & retention (`OneDrive Storage Quota – Default – Prod`, `OneDrive Retention – Orphaned – Prod`)

Procedure:

```
Get-SPOTenant | Select OneDriveStorageQuota, OrphanedPersonalSitesRetentionPeriod
```

Expected pass: both equal the approved design values (e.g. quota 1048576 MB, retention 365 days); a per-user override (if set) shows on that site via `Get-SPOSite -Identity <url> | Select StorageQuota`. **Fail:** values differ from design — re-run `ODFBGOV-Configure.ps1 -Commit`.

Test 6 — Sharing alignment (`OneDrive Sharing – Aligned – Prod`)

Procedure:

```
Get-SPOTenant | Select SharingCapability, OneDriveSharingCapability
```

Expected pass: `OneDriveSharingCapability` equals the target and is **no more permissive** than the org `SharingCapability` (e.g. org `ExternalUserSharingOnly`, OneDrive `ExistingExternalUserSharingOnly`). **Fail:** OneDrive more permissive than org (not possible to set — indicates a misread target) or not the approved value — re-align.

Test 7 — Offboarding access (`Offboarding Access – Manager – Prod`)

Procedure: For a **test** departing user, run `ODFBGOV-Set-OffboardingAccess.ps1 … -Commit`, then have the delegate open the leaver's OneDrive URL.

```
Get-SPOUser -Site <OneDriveUrl> | Where-Object { $_.LoginName -eq 'manager@contoso.com' } | Select LoginNa
```

Expected pass: delegate shows `IsSiteAdmin = True`, can browse and download the leaver's files; the reported retention window matches `OneDrive Retention – Orphaned – Prod`. **Fail:** access denied or OneDrive not resolved — see Troubleshooting (offboarding access).

Test 8 — Governance report (`OneDrive Governance – Tenant – Report`)

Procedure:

```
.\ODFBGOV-Report.ps1 -AdminUrl https://contoso-admin.sharepoint.com -HtmlPath C:\ODFBGOV\OneDrive-Report.h
```

Expected pass: the report renders one row per OneDrive with quota, used, headroom %, last activity, sharing, and NearQuota/Stale flags; totals match the provisioned OneDrive count. **Fail:** empty/partial report — confirm Graph `Reports.Read.All` consent and that usage reporting is not de-identified (a tenant privacy setting can anonymize names).

Go / No-Go gate (before Stage 8 expansion)

Go only when Tests 1–8 all pass on the pilot ring: silent sign-in works, KFM redirects and blocks opt-out, sync is restricted to allowed devices and blocks personal accounts, quota/retention/sharing match design, offboarding grants work, and the governance report is clean. **No-Go on any failure** — remediate via Troubleshooting and re-test before expanding tenant-wide.

6. Troubleshooting — OneDrive for Business Deployment & Governance

Common failure modes, their root cause, and the resolution. Symptoms reference the named artifacts and tests from the Implementation and Testing guides.

Silent sign-in doesn't happen (client shows "Sign in")

Symptom: After `OneDrive Client Config – Pilot – Stage1` applies, the OneDrive client still prompts to sign in; Test 1 shows `SilentAccountConfig` present but the account isn't configured. **Root causes & resolution:**

- **Device isn't Entra/hybrid joined.** `SilentAccountConfig` signs in with the device's primary work account; on workgroup/personal devices it cannot complete. Confirm `dsregcmd /status` shows `AzureAdJoined : YES` (Requirement 2). Resolution: bring the device into Entra/hybrid join.
- **Policy not applied.** Confirm the profile is assigned to `OneDrive Pilot Ring – Users – Stage1` and the device has synced policy (Intune → device → check-in; or `gpupdate /force` for GPO). Verify `HKLM\SOFTWARE\Policies\Microsoft\OneDrive\SilentAccountConfig = 1`.
- **Modern auth / MFA loop.** If the account requires interactive MFA at first sign-in, silent config completes only after the credential is established; ensure the device is compliant for conditional access.

KFM didn't redirect (folders still on the local profile)

Symptom: Desktop/Documents/Pictures still resolve to `%UserProfile%\...` not `…\OneDrive - <Org>\`; Test 2 fails. **Root causes & resolution:**

- **Wrong or blank tenant ID.** `KFMSilentOptIn` must equal your **tenant ID** GUID (Step 0.2). A wrong or empty value silently no-ops. Fix the value (use `ODFBGOV-Set-Scope.ps1 -TenantOnlyLookup`) and reapply.
- **Folder ineligible.** Legacy **Folder Redirection** (GPO) pointing a known folder to a network/UNC path, a known folder on a non-C: drive, or a folder containing OneNote notebooks/locked or unsupported files can block the silent move. Remediation: remove the conflicting Folder Redirection GPO, then let KFM retry. Files already there stay; the move completes once the blocker is cleared.
- **Not signed in yet.** KFM runs after the client is signed in; if silent sign-in failed (above), fix that first — KFM has nothing to redirect into.

Sync isn't blocked on an unmanaged device (restriction not working)

Symptom: A non-domain-joined device still syncs; Test 3 fails. **Root causes & resolution:**

- **Restriction not enabled / wrong GUID.** `Get-SPOTenantSyncClientRestriction` must show `TenantRestrictionEnabled = True` with the correct **domain GUID** (not the tenant GUID — the domain-join list uses the AD **domain** GUID from `(Get-ADDomain).ObjectGUID`). Re-apply with the correct GUID via `ODFBGOV-Configure.ps1 -DomainGuids … -Commit`.
- **Propagation delay.** Sync-client restriction changes can take time to take effect on clients; allow the documented propagation window before concluding it failed, and re-test on a clean client.

Sync is blocked on a device that should be allowed (over-blocked)

Symptom: A legitimately domain-joined device gets "your organization does not allow syncing on this computer." **Root causes & resolution:**

- **Device in a different domain/forest.** Only the domain GUID(s) in the allow-list may sync. Add the missing domain GUID (multi-domain forests need each domain's GUID) and reapply.
- **Hybrid-only device without the on-prem domain GUID.** Devices joined only to Entra (not the on-prem AD) won't match an on-prem domain GUID. Reconsider whether domain-join restriction fits your device model, or use device compliance/conditional access instead — **verify which control matches your estate.**

Personal account still syncs (personal-sync block not working)

Symptom: Users can add a personal Microsoft account; Test 4 fails. **Resolution:** Confirm `DisablePersonalSync = 1` is in the assigned profile and applied on the device (`HKLM\SOFTWARE\Policies\Microsoft\OneDrive\DisablePersonalSync`). If present but ineffective, ensure the OneDrive client build supports the policy and re-sync policy.

Offboarding — can't access the departed user's OneDrive

Symptom: `ODFBGOV-Set-OffboardingAccess.ps1` reports "No provisioned OneDrive found," or the delegate is denied. **Root causes & resolution:**

- **OneDrive never provisioned.** A user who never signed in to OneDrive may have no personal site; it provisions on first use (or can be pre-provisioned with `Request-SPOPersonalSite`). If never provisioned, there is no data to hand off.
- **Owner UPN mismatch.** The script matches on `Owner`; if the UPN changed (rename/merge), pass the current UPN or locate the site with `Get-SPOSite -IncludePersonalSite $true -Filter "Url -like '…'"`.
- **Retention already expired.** If the account was deleted more than `OrphanedPersonalSitesRetentionPeriod`

days ago, the OneDrive is permanently removed. This is why the offboarding grant must happen **before** deletion, and holds must be confirmed with compliance first.

Governance report is empty or partial

Symptom: `ODFBGOV-Report.ps1` returns few/no rows or blank last-activity. **Root causes & resolution:**

- **Graph consent missing.** The report needs `Reports.Read.All`; consent it on first run.
- **Usage reports de-identified.** A tenant privacy setting (M365 admin center → Settings → Org settings → Reports) anonymizes user names in usage reports, breaking per-user matching. Turn it off (with privacy approval) to get named data; otherwise storage/quota still report from SPO but activity won't match.
- **Latency.** Usage data lags actual activity by a few days and has a fixed maximum window — set `-UsagePeriod` inside the available range.

`Connect-SPOService` fails (wrong admin URL / cloud)

Symptom: Connection error against the admin endpoint. **Resolution:** Use the correct admin URL for your cloud — Commercial `https://<tenant>-admin.sharepoint.com`, Government `https://<tenant>-admin.sharepoint.us` (**verify in your tenant**). Ensure the account holds the least-privilege SharePoint Administrator role and modern auth/MFA succeeds.

7. Best Practices — OneDrive for Business Deployment & Governance

Durable practices that keep OneDrive deployed, governed, and low-risk after the initial rollout.

Deployment & client

- **Silent sign-in + KFM together, always.** Silent sign-in without KFM leaves Desktop/Documents/Pictures on the device — the very data-loss gap you deployed to close. Pair `SilentAccountConfig` with `KFMSilentOptIn` in the same profile so protection is complete the moment the client signs in.
- **Block the opt-out.** Set `KFMBlockOptOut = 1` so users can't move known folders back to the PC and silently re-open the data-loss gap. Use the notification variant so users understand the change.
- **Keep the client current, don't repackage it.** OneDrive updates itself on its release ring; prefer the built-in current ring over pinning an old standalone build. Only use per-machine standalone install where your device model genuinely requires it (e.g. non-persistent VDI).
- **Enable Files On-Demand.** `FilesOnDemandEnabled = 1` avoids filling local disks on first sync of a large OneDrive — essential for KFM at scale.

Sync governance

- **Restrict sync to managed devices.** Always enable the domain-join allow-list

(`Sync Restriction – DomainJoined – Prod`) so corporate data only syncs to devices you manage, and block personal accounts (`DisablePersonalSync`) so corporate and personal data never mix on a device.

- **Block risky file types at the tenant.** Exclude extensions that should never sync (e.g. `pst`) via

`Set-SPOTenantSyncClientRestriction -ExcludedFileExtensions`.

- **Use web-only access for unmanaged sessions** (`ConditionalAccessPolicy = AllowLimitedAccess`) so

browser access from unmanaged devices can't download, print, or sync — verify the licensing first.

- **Prefer ADMX/Intune over hand-edited registry.** Deploy client policies through the OneDrive ADMX in

Intune/GPO, not by pushing raw registry values — the ADMX carries the correct, build-matched value names.

Storage, retention & sharing

- **Set a deliberate default quota, override by exception.** A governed default

(`OneDrive Storage Quota – Default – Prod`) with per-role overrides beats an unbounded max that hides cost and hoards stale data. Review near-quota accounts from the governance report.

- **Set the orphaned-retention window with records, not by default.** It is the **hard deletion timer**

for a departed user's OneDrive — agree it with compliance/records and document it.

- **Keep OneDrive sharing aligned to (≤) org sharing.** Re-check `OneDriveSharingCapability` against org

`SharingCapability` after any sharing change so OneDrive never drifts more permissive than policy.

Offboarding & lifecycle

- **Grant, retrieve, then revoke.** Grant the manager site-collection admin only for the handoff window

and remove it (`IsSiteCollectionAdmin $false`) once files are retrieved — don't leave standing access.

- **Confirm holds before deletion.** Never delete a departed user's account until compliance confirms no

legal/records hold applies — the retention window is a timer, not a hold, and not a backup.

- **Automate the manager-access grant** on deletion (SharePoint admin center Retention/Notifications) so

offboarding is consistent, not ad-hoc.

Operate & assure

- **Report on a cadence.** Run `ODFBGOV-Report.ps1` regularly and act on NearQuota (≥90%) and Stale

accounts; a downward stale trend means governance is working.

- **Pilot every high-impact change.** Sync, sharing, retention, and KFM changes are tenant-wide — always

validate on `OneDrive Pilot Ring – Users – Stage1` with the go/no-go gate before broad rollout, and keep the documented rollback ready.

- **Preview before you commit.** Run `ODFBGOV-Configure.ps1` and `ODFBGOV-Set-OffboardingAccess.ps1`

without `-Commit` (or with `-WhatIf`) first; commit only after the change table is reviewed.

- **Verify plan- and cloud-dependent behavior in your tenant.** Storage maxima, some sharing controls,

and admin URLs differ by plan and between Commercial and Government clouds — confirm, don't assume.

8. Deliverables Summary — OneDrive for Business Deployment & Governance

Everything included in this solution, on one page.

Documentation

Document	What it gives you
01 – Solution Overview	The OneDrive model: silent client deployment, KFM, sync controls, storage/retention, sharing alignment, and offboarding — plus the reference design and the artifact naming convention.
02 – Requirements	The six OneDrive-specific prerequisites (managed-device policy channel, Entra join for silent sign-in, tenant ID + domain GUIDs, storage/retention decision, org sharing ceiling, offboarding model) — each with Verify / Implement — building on the SharePoint category global requirements.
03 – Implementation	Exact, ordered, pilot-first steps: Assess → Design → Configure (client, KFM, sync, storage/retention/sharing, offboarding) → Pilot → Validate → Expand, with pass/fail gates and a documented rollback. Every artifact named.
05 – Testing & Validation	Eight validation tests (silent sign-in, KFM, sync restriction, personal-sync block, quota/retention, sharing alignment, offboarding access, governance report), each with expected pass/fail, and the go/no-go gate.

Document	What it gives you
06 – Troubleshooting	The common failure modes (silent sign-in not working, KFM blocked, sync not blocked/over-blocked, offboarding access, report empty) with root cause and resolution.
07 – Best Practices	Durable practices for client, sync, storage/retention/sharing, offboarding, and ongoing assurance.
Executive Summary	The outcome, approach, and why the rollout is low-risk — for approvers.

PowerShell scripts (5) — safe by default (read-only or `-WhatIf`)

Script	What it does
ODFBGOV-Discovery.ps1	**Read-only.** Assesses the OneDrive posture: tenant default quota, orphaned-site retention, OneDrive sharing capability, conditional/limited access, sync-client restriction, provisioned OneDrive sites, and (optional) Graph usage — writes the `OneDrive Baseline – Tenant – Discovery` export.
ODFBGOV-Configure.ps1	**Dry-run by default (`-WhatIf` / no `-Commit`).** Tenant sync-control + storage/retention/sharing helper: sets default quota, orphaned retention, `OneDriveSharingCapability`, web-only access, and the domain-join sync allow-list — previews every change (current → proposed) before applying with `-Commit`.
ODFBGOV-Set-OffboardingAccess.ps1	**Dry-run by default.** Offboarding/data-retention helper: resolves a departing user's OneDrive, grants the manager site-collection admin for handoff, and reports the retention window — never deletes anything; legal/records hold deferred to compliance.
ODFBGOV-Report.ps1	**Read-only.** Governance report: per-OneDrive storage used vs quota (headroom %), last activity, sharing, and NearQuota/Stale flags, to CSV/HTML — the `OneDrive Governance – Tenant – Report`.
ODFBGOV-Set-Scope.ps1	**Read-only helper.** Resolves the tenant ID (KFM + tenant sync restriction) and AD domain GUID(s) (domain-join sync restriction), and emits a reviewed `.reg` reference template of the documented client policy value names.

All scripts use modern-auth `Connect-*` (no stored credentials or secrets), default to read-only or `-WhatIf`, and require `-Commit` to write. Cmdlet/parameter and portal detail follow Microsoft public documentation; plan- and cloud-dependent behavior (storage maxima, Government cloud URLs) is flagged **"verify in your tenant."**

Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

ODFBGOV-Configure.ps1

Purpose: Applies the tenant-level OneDrive governance posture in one safe, repeatable pass: default storage quota, orphaned-OneDrive retention period, OneDrive sharing capability (kept no more permissive than the org sharing level), optional web-only/limited access (blocks download and sync in the browser), and the sync-client domain-join allow list (domain GUIDs). SAFE BY DEFAULT: with no -Commit it only PREVIEWS every change (current -> proposed) and writes nothing. HOW TO RUN Preview (no changes): .\ODFBGOV-Configure.ps1 -AdminUrl https://contoso-admin.sharepoint.com -StorageQuotaMB 1048576 -RetentionDays 365 -OneDriveSharing ExistingExternalUserSharingOnly -DomainGuids '12345678-90ab-cdef-1234-567890abcdef' -WhatIf Apply: .\ODFBGOV-Configure.ps1 -AdminUrl https://contoso-admin.sharepoint.com -StorageQuotaMB 1048576 -RetentionDays 365 -Commit (PowerShell 5.1+/7+; Microsoft.Online.SharePoint.PowerShell; Connect-SPOService modern auth. Get domain GUIDs from ODFBGOV-Set-Scope.ps1. Government cloud admin URL is *-admin.sharepoint.us — verify.) EXPECTED RESULTS A change table (setting / current / proposed / action). With -Commit the named artifacts "OneDrive Storage Quota - Default - Prod", "OneDrive Retention - Orphaned - Prod", "OneDrive Sharing - Aligned - Prod", and "Sync Restriction - DomainJoined - Prod" are applied. .NOTES SharePoint category — OneDrive for Business Deployment & Governance. Writes only with -Commit; honors -WhatIf. Tenant-wide sharing/retention/sync changes are HIGH-IMPACT — run under change control, pilot-first. > [CmdletBinding(SupportsShouldProcess, ConfirmImpact = 'High')] param([Parameter(Mandatory)][string]$AdminUrl, [int]$StorageQuotaMB, [ValidateRange(30,3650)][int]$RetentionDays, [ValidateSet('Disabled','ExistingExternalUserSharingOnly','ExternalUserSharingOnly','ExternalUserAndGuestSharing')] [string]$OneDriveSharing, [ValidateSet('AllowFullAccess','AllowLimitedAccess')][string]$ConditionalAccess, [string[]]$DomainGuids, [switch]$Commit)

How to run: Preview (no changes): .\ODFBGOV-Configure.ps1 -AdminUrl https://contoso-admin.sharepoint.com -StorageQuotaMB 1048576 -RetentionDays 365 -OneDriveSharing ExistingExternalUserSharingOnly -DomainGuids '12345678-90ab-cdef-1234-567890abcdef' -WhatIf Apply: .\ODFBGOV-Configure.ps1 -AdminUrl https://contoso-admin.sharepoint.com -StorageQuotaMB 1048576 -RetentionDays 365 -Commit (PowerShell 5.1+/7+; Microsoft.Online.SharePoint.PowerShell; Connect-SPOService modern auth. Get domain GUIDs from ODFBGOV-Set-Scope.ps1. Government cloud admin URL is *-admin.sharepoint.us — verify.) EXPECTED RESULTS A change table (setting / current / proposed / action). With -Commit the named artifacts "OneDrive Storage Quota - Default - Prod", "OneDrive Retention - Orphaned - Prod", "OneDrive Sharing - Aligned - Prod", and "Sync Restriction - DomainJoined - Prod" are applied. .NOTES SharePoint category — OneDrive for Business Deployment & Governance. Writes only with -Commit; honors -WhatIf. Tenant-wide sharing/retention/sync changes are HIGH-IMPACT — run under change control, pilot-first. > [CmdletBinding(SupportsShouldProcess, ConfirmImpact = 'High')] param([Parameter(Mandatory)][string]$AdminUrl, [int]$StorageQuotaMB, [ValidateRange(30,3650)][int]$RetentionDays, [ValidateSet('Disabled','ExistingExternalUserSharingOnly','ExternalUserSharingOnly','ExternalUserAndGuestSharing')] [string]$OneDriveSharing, [ValidateSet('AllowFullAccess','AllowLimitedAccess')][string]$ConditionalAccess, [string[]]$DomainGuids, [switch]$Commit)

ODFBGOV-Discovery.ps1

Purpose: Builds the OneDrive governance baseline before any change. Reads the tenant-level OneDrive settings from the SharePoint Online Management Shell (default storage quota, orphaned-site retention, OneDrive sharing capability, conditional-access/limited-access, sync-client restriction), enumerates provisioned OneDrive personal sites, and (optionally) pulls the Microsoft Graph OneDrive usage report to show storage used and last activity per account. Read-only — makes NO changes. HOW TO RUN .\ODFBGOV-Discovery.ps1 -AdminUrl https://contoso-admin.sharepoint.com -CsvPath C:\Temp\ODFBGOV-Baseline.csv (PowerShell 5.1+/7+; Microsoft.Online.SharePoint.PowerShell + optionally Microsoft.Graph. Connect with Connect-SPOService — modern auth / MFA. For Government clouds the admin URL is -admin.sharepoint.us — verify in your tenant.) EXPECTED RESULTS A console summary of the tenant OneDrive settings and a per-OneDrive CSV (owner, URL, quota, storage used, last activity). No writes occur; this is the artifact "OneDrive Baseline - Tenant - Discovery". .NOTES SharePoint category — OneDrive for Business Deployment & Governance. Read-only against SPO + Graph. > [CmdletBinding()] param([Parameter(Mandatory)][string]$AdminUrl, [string]$CsvPath, [switch]$IncludeGraphUsage, [ValidateSet('D7','D30','D90','D180')][string]$UsagePeriod = 'D30')

How to run: .\ODFBGOV-Discovery.ps1 -AdminUrl https://contoso-admin.sharepoint.com -CsvPath C:\Temp\ODFBGOV-Baseline.csv (PowerShell 5.1+/7+; Microsoft.Online.SharePoint.PowerShell + optionally Microsoft.Graph. Connect with Connect-SPOService — modern auth / MFA. For Government clouds the admin URL is -admin.sharepoint.us — verify in your tenant.) EXPECTED RESULTS A console summary of the tenant OneDrive settings and a per-OneDrive CSV (owner, URL, quota, storage used, last activity). No writes occur; this is the artifact "OneDrive Baseline - Tenant - Discovery". .NOTES SharePoint category — OneDrive for Business Deployment & Governance. Read-only against SPO + Graph. > [CmdletBinding()] param([Parameter(Mandatory)][string]$AdminUrl, [string]$CsvPath, [switch]$IncludeGraphUsage, [ValidateSet('D7','D30','D90','D180')][string]$UsagePeriod = 'D30')

ODFBGOV-Report.ps1

Purpose: Produces the recurring governance report leadership and admins review: per-OneDrive storage used vs quota (headroom %), last activity date, external sharing capability, and flags for near-quota (>=90%) and stale (no activity in the window) accounts. Combines the SharePoint Online site inventory with the Microsoft Graph OneDrive usage report. Read-only — no changes. HOW TO RUN .\ODFBGOV-Report.ps1 -AdminUrl https://contoso-admin.sharepoint.com -HtmlPath C:\Temp\ODFBGOV-Report.html -StaleDays 90 (PowerShell 5.1+/7+; Microsoft.Online.SharePoint.PowerShell + Microsoft.Graph. Connect-SPOService modern auth; Graph scope Reports.Read.All.) EXPECTED RESULTS Console top-N tables plus an HTML/CSV artifact "OneDrive Governance - Tenant - Report": one row per OneDrive with quota, used, headroom %, last activity, sharing, and NearQuota/Stale flags. .NOTES SharePoint category — OneDrive for Business Deployment & Governance. Read-only against SPO + Graph. > [CmdletBinding()] param([Parameter(Mandatory)][string]$AdminUrl, [string]$HtmlPath, [string]$CsvPath, [int]$StaleDays = 90, [ValidateSet('D7','D30','D90','D180')][string]$UsagePeriod = 'D90')

How to run: .\ODFBGOV-Report.ps1 -AdminUrl https://contoso-admin.sharepoint.com -HtmlPath C:\Temp\ODFBGOV-Report.html -StaleDays 90 (PowerShell 5.1+/7+; Microsoft.Online.SharePoint.PowerShell + Microsoft.Graph. Connect-SPOService modern auth; Graph scope Reports.Read.All.) EXPECTED RESULTS Console top-N tables plus an HTML/CSV artifact "OneDrive Governance - Tenant - Report": one row per OneDrive with quota, used, headroom %, last activity, sharing, and NearQuota/Stale flags. .NOTES SharePoint category — OneDrive for Business Deployment & Governance. Read-only against SPO + Graph. > [CmdletBinding()] param([Parameter(Mandatory)][string]$AdminUrl, [string]$HtmlPath, [string]$CsvPath, [int]$StaleDays = 90,

[ValidateSet('D7','D30','D90','D180')][string]$UsagePeriod = 'D90')

ODFBGOV-Set-OffboardingAccess.ps1

Purpose: On a user departure, preserves and hands off their OneDrive data WITHOUT deleting anything. Resolves the user's OneDrive personal site, grants a delegate (typically the manager) as a site-collection administrator so they can retrieve needed files, and reports the tenant orphaned-OneDrive retention period so operators know how long the data survives account deletion. SAFE BY DEFAULT: previews the grant unless -Commit is passed. It never deletes the account, the OneDrive, or any content, and never removes a hold. HOW TO RUN Preview: .\ODFBGOV-Set-OffboardingAccess.ps1 -AdminUrl https://contoso-admin.sharepoint.com -DepartingUser jdoe@contoso.com -Delegate manager@contoso.com -WhatIf Apply: .\ODFBGOV-Set-OffboardingAccess.ps1 -AdminUrl https://contoso-admin.sharepoint.com -DepartingUser jdoe@contoso.com -Delegate manager@contoso.com -Commit (PowerShell 5.1+/7+; Microsoft.Online.SharePoint.PowerShell; Connect-SPOService modern auth.) EXPECTED RESULTS Confirms the resolved OneDrive URL, the retention period in days, and (with -Commit) that the delegate now holds site-collection admin on "Offboarding Access - Manager - Prod". Legal hold / records determinations are DEFERRED to compliance — this only grants operational access. .NOTES SharePoint category — OneDrive for Business Deployment & Governance. Writes only with -Commit; honors -WhatIf. Placing the OneDrive on retention/eDiscovery hold (Purview) is a COMPLIANCE decision — out of scope here. > [CmdletBinding(SupportsShouldProcess, ConfirmImpact = 'High')] param([Parameter(Mandatory)][string]$AdminUrl, [Parameter(Mandatory)][string]$DepartingUser, [Parameter(Mandatory)][string]$Delegate, [switch]$Commit)

How to run: Preview: .\ODFBGOV-Set-OffboardingAccess.ps1 -AdminUrl https://contoso-admin.sharepoint.com -DepartingUser jdoe@contoso.com -Delegate manager@contoso.com -WhatIf Apply: .\ODFBGOV-Set-OffboardingAccess.ps1 -AdminUrl https://contoso-admin.sharepoint.com -DepartingUser jdoe@contoso.com -Delegate manager@contoso.com -Commit (PowerShell 5.1+/7+; Microsoft.Online.SharePoint.PowerShell; Connect-SPOService modern auth.) EXPECTED RESULTS Confirms the resolved OneDrive URL, the retention period in days, and (with -Commit) that the delegate now holds site-collection admin on "Offboarding Access - Manager - Prod". Legal hold / records determinations are DEFERRED to compliance — this only grants operational access. .NOTES SharePoint category — OneDrive for Business Deployment & Governance. Writes only with -Commit; honors -WhatIf. Placing the OneDrive on retention/eDiscovery hold (Purview) is a COMPLIANCE decision — out of scope here. > [CmdletBinding(SupportsShouldProcess, ConfirmImpact = 'High')] param([Parameter(Mandatory)][string]$AdminUrl, [Parameter(Mandatory)][string]$DepartingUser, [Parameter(Mandatory)][string]$Delegate, [switch]$Commit)

ODFBGOV-Set-Scope.ps1

Purpose: Removes the guesswork from OneDrive client policy by producing the exact values operators must paste into Intune / Group Policy and the SharePoint admin center: the Microsoft 365 tenant ID (needed for the KFM "Silently move known folders" policy and for tenant sync restriction) and the on-prem Active Directory domain GUID(s) (needed for "Allow syncing only on computers joined to specific domains"). Optionally emits a ready-to-import .reg reference of the documented OneDrive client policy value names. Read-only — resolves and reports, changes nothing. HOW TO RUN .\ODFBGOV-Set-Scope.ps1 -TenantOnlyLookup # tenant ID via Graph .\ODFBGOV-Set-Scope.ps1 -ResolveDomainGuids # AD domain GUIDs (needs RSAT ActiveDirectory) .\ODFBGOV-Set-Scope.ps1

-TenantOnlyLookup -EmitRegReference C:\Temp\ODFBGOV-ClientPolicy.reg (PowerShell 5.1+/7+; Microsoft.Graph for tenant ID; ActiveDirectory module for domain GUIDs.) EXPECTED RESULTS Prints the tenant ID and each domain's Name + ObjectGUID, and (with -EmitRegReference) writes a commented .reg TEMPLATE of the documented policy value names (SilentAccountConfig, KFMSilentOptIn, KFMBlockOptOut, FilesOnDemandEnabled) with the tenant GUID substituted — a REFERENCE to review, not an auto-apply. Verify value names against the current OneDrive.admx for your build. .NOTES SharePoint category — OneDrive for Business Deployment & Governance. Read-only; the .reg output is a template to review, never silently imported. > [CmdletBinding()] param([switch]$TenantOnlyLookup, [switch]$ResolveDomainGuids, [string]$EmitRegReference)

How to run: .\ODFBGOV-Set-Scope.ps1 -TenantOnlyLookup # tenant ID via Graph .\ODFBGOV-Set-Scope.ps1 -ResolveDomainGuids # AD domain GUIDs (needs RSAT ActiveDirectory) .\ODFBGOV-Set-Scope.ps1 -TenantOnlyLookup -EmitRegReference C:\Temp\ODFBGOV-ClientPolicy.reg (PowerShell 5.1+/7+; Microsoft.Graph for tenant ID; ActiveDirectory module for domain GUIDs.) EXPECTED RESULTS Prints the tenant ID and each domain's Name + ObjectGUID, and (with -EmitRegReference) writes a commented .reg TEMPLATE of the documented policy value names (SilentAccountConfig, KFMSilentOptIn, KFMBlockOptOut, FilesOnDemandEnabled) with the tenant GUID substituted — a REFERENCE to review, not an auto-apply. Verify value names against the current OneDrive.admx for your build. .NOTES SharePoint category — OneDrive for Business Deployment & Governance. Read-only; the .reg output is a template to review, never silently imported. > [CmdletBinding()] param([switch]$TenantOnlyLookup, [switch]$ResolveDomainGuids, [string]$EmitRegReference)